

SEBA Master: Web Application Engineering

0. Introduction

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After successful participation in this lecture, you can

- identify, explain and develop B2B & B2C **business models** on the web
 - select and apply **patterns** for web application design
 - explain a **user-centered, iterative development process** and apply it in a team
 - explain and apply concepts of **modern web technologies**
 - **present** software projects and software deliverables to an expert audience
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- **apply** concepts and technologies of a **JavaScript-based web application development framework** to build a small web application
 - **develop** small web applications based on the **MERN** stack (**M**ongoDB, **E**xpress, **R**ect, **N**odeJS)
 - **containerize** your web application locally using **Docker**
 - have sufficient knowledge to contribute to the development of projects in this area

Outline of the Lecture (1)

1. Web Site Genres

- Business Models on the Web
- Business Model Canvas & Value Proposition Canvas

2. Web Site Design Process with Patterns

- Key Issues of Customer-Centered Web Design
- Knowing Your Customers
- Involving Customers with Iterative Design
- Additional Viewpoints and Aspects
- **Appendix: Web Site Design Patterns**

3. Developing Single-Page Web Applications

- Introduction: Architecture of Web Applications
- HTML & CSS Concepts
- JavaScript Basics
- Asynchronous Operations in JavaScript
- Single-Page Applications
- From Mockups to Web Components

4. Developing Single-Page Web Applications with React

- Web Application Framework Comparison
- Basic Concepts of React
- State Management
- Navigation & Routing
- Client-Server Communication
- Consuming RESTful Services

Outline of the Lecture (3)

5. Building REST-Enabled Backend Services

- Target Architecture and Development Environment
- Event-driven Architecture and Asynchronous Operations with NodeJS
- Creating REST Interfaces Using Express
- Using Document-oriented Database Storage: MongoDB Example
- Enabling User Authentication on the Web Service Using JSON Web Tokens (JWT)

6. Advanced Topics in Web Application Engineering

- Real-time Web Applications
- Hybrid Web Applications

7. High Performance Web Applications

- Database Options for Web Applications
- Scaling Web Applications

Guest Lecture

- TBD

Organizational Issues – Common Information

- **Module number:** IN2087
- **ECTS:** 8 Credits
- **Workload:** 240 hours
- **Target audience**
 - Master Information Systems (compulsory course, priority)
 - ~~Master Informatics & others (elective course)~~
- **Prerequisites**
 - Bachelor-level knowledge of programming, software engineering, networks, and databases.
- **Lecture materials**
 - Slides and assignments will be available on [Moodle](#).
 - To get access to the course on Moodle, you need to be registered to the course on [TUMonline](#).
- **Type of Assessment: Project Work**
 - Development of a web application in a team of 4 students.
 - Assessment based on (written) deliverables and presentations.

- **Course Format:**
 - The lecture is divided into a theoretical and a practical part.
First central exercise on **April 22nd**.
 - **Lecture (theoretical part): Mondays 14:15 – 15:45 (onsite)**
 - **Central Exercise (practical part): certain Wednesdays 14:15 – 15:45 (Zoom)**
 - See lecture dates: [SEBA Master Homepage](#)
- **Submission of deliverables and group presentations (Each part contributes to the final grade.):**
 1. Business idea, BMC, and VPC (Submission: 03.05.26 – Presentations: 28.04. - 01.05.)
 2. Customer journey, mockups, and data model (Submission: 17.05.26 - Presentations: 19.05. - 22.05.)
 3. Final Prototype (Submission: 12.07.26 - Presentations: 20.07. – 24.07.)
- **Special guest** who talks about **web application engineering** in the **industry** – **TBD**
 - Topic: TBD
 - Date: TBD

- **The registration for the course is over!**
- Places are limited to 228 due to capacity constraints (57 teams of 4 students).
- In the first week of the lecture, students **with fixed places** will be **assigned to teams** by the organizer.
- In case you want to form a **team on your own**, please do so on seba-master.sebis.cit.tum.de.
 - You should have received an invitation E-Mail already
 - **Deadline: 20.04.2026 at 23:59 (TODAY)**
- If you are no longer willing to participate in the course, please deregister on TUMonline **and write us a short email** so that your place can be given to another student (only possible in the first week).

- **Each team** will be assigned to a **mentor** (employees/research assistants of the sebis chair).
- **Mentors** will provide support by advising you throughout the course. Feel free to contact them.
- **Please note:**
 1. They are not your "tech support". Please try to solve technical problems within the team and only reach out to the mentors regarding such issues if you can't solve the problem on your own.
 2. Don't ask the mentors to grade your submissions in advance. If you want feedback on your submission before the submission deadline, please contact your mentor **early**, i.e., a few days in advance.
- When contacting your mentor **via mail**, please mention your **team number** and put the remaining **team members in CC**.
- Getting **early** feedback from the advisor, especially for A1 and A2, is **highly recommended** (and mandatory for A1)!

Details on the Overall Assessment

The final grade consists of **three partial assessments**:

- Over the course of the semester, we will have **three assignments**
 - **A1** – Business idea evaluation: Business idea, BMC, and VPC (10%)
 - **A2** – Technical concept: Customer journey, mockups, and data model (15%)
 - **A3** – Final prototype (75%)
- Individual assessment regarding **presentation skills, rhetoric**, and performance in **discussion**

Assessment	Contribution	Example
A1 – Business Idea Evaluation	10%	1.0
A2 – Technical Concept	15%	1.7
A3 – Final Prototype	75%	1.3
	Final Grade	1.3

- **However:** In each assignment we assess the individual contribution of each team member. Should a team member not contribute to the group work, we will assign that team member an overall final grade of 5.0, which results in failing the course regardless of potentially already acquired partial assessments of ≥ 4.0 .

Finding Your Idea

Call for sophisticated project ideas by experienced students

If you already have some **experience** in web application development, do not hesitate to approach more **sophisticated topics**, e.g., by implementing/integrating

- Real-time collaboration
- Interactive visualizations and maps
- Interactive video players & augmented reality
- Machine learning, natural language processing, etc.
- ...

Successfully implementing **ambitious** and **innovative** web applications can

- have a positive impact on the final grading of your project (we would ignore certain technical issues if the innovative core functionality is implemented satisfactorily)
- potentially lead to interesting follow-up projects (e.g., master's thesis, cooperation with industry partners, etc.)
- be an interesting challenge for you

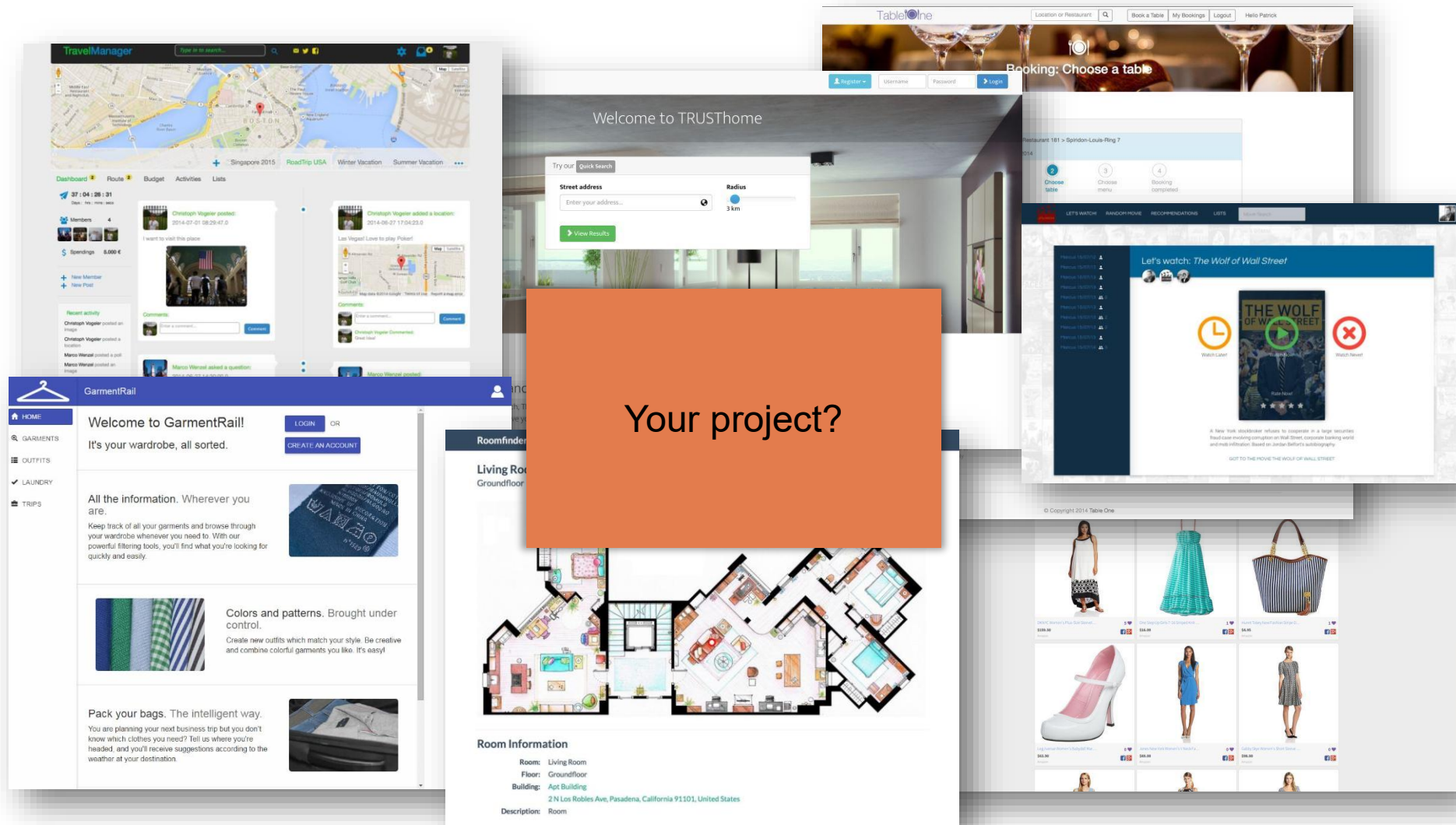
Currently, there are some **related research projects** at the sebis chair to which your idea could contribute, e.g., *NLP & LegalTech*, *Next-Generation IT Organizations*, *Service Platform and Ecosystems*, *Blockchain-based Systems Engineering*, and more. (Check the [sebis page](#) for further details.)

However, **do not overreach yourself**, especially if you do not have any experience in web application development.

Finding Your Idea

Projects from previous years

Over the last few years, we have published a selection of great student projects on [our website](#).

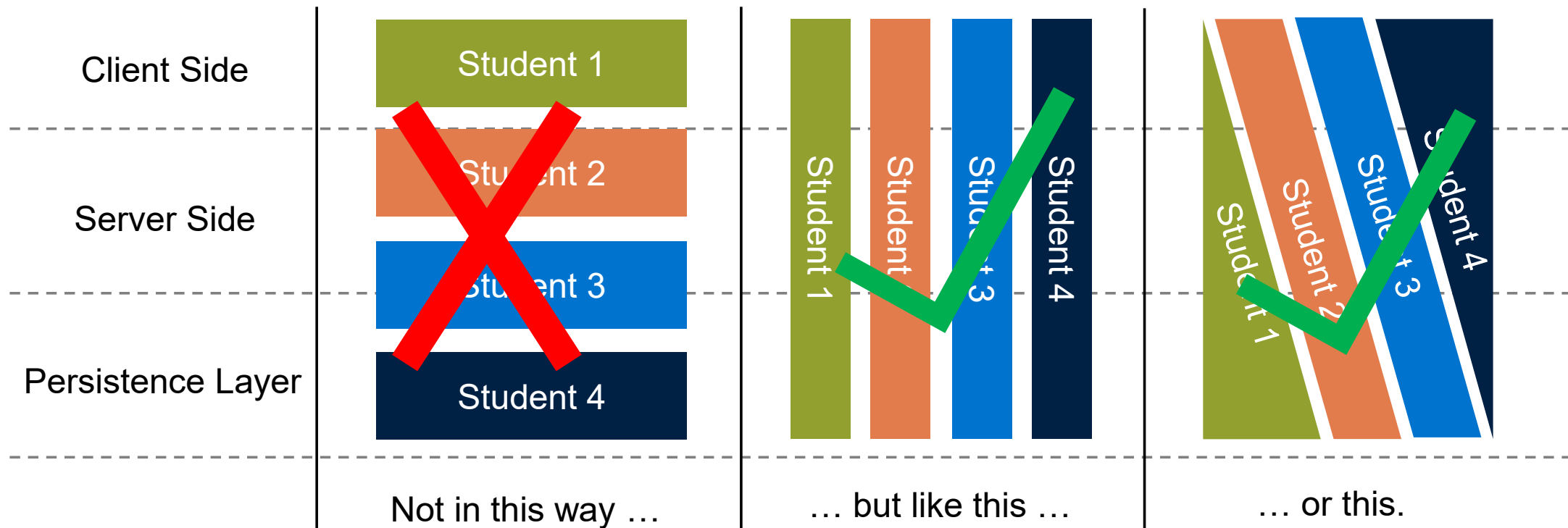


Structure of the Presentations

- Over the course of the semester, we will have **1 feedback and 2 presentation blocks**.
- For the feedback block, you need to schedule the appointment yourselves (check Assignment 1).
- Each presentation block lasts **4-5 days** and has approx. 10-12 slots in total (~3 slots per day).
- We will assign each student team to one of these slots. You will be able to provide your availability, and we will respect it as good as mathematically possible.
- Each team must be **present for the entire duration of its assigned slot**.
- In every slot there will be 4-5 student teams.
- The presentations will be performed **onsite**.
 - Each team of a certain slot will receive a calendar invitation for this slot.
 - The **whole team joins the meeting at the start time and stays until the end** of the last presentation in that slot.
 - The schedule for the 3 presentation blocks will be published roughly a week before the respective block takes place.

Distribution of Workload

- Distribute the workload of each assignment in a way that each student is familiar with each part of the deliverable, not only with those he/she is responsible for.
 - Example: For the implementation of the prototype, do not distribute the workload horizontally along the abstraction layers, but vertically along the use cases.
 - In the discussion we might ask any student about any part of the deliverables.



Technology Stack (MERN)

